PERFECTNESS WITH TOR AMPLITUDE IN A RANGE CAN BE CHECKED ON RESIDUE FIELDS AT MAXIMAL IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof span opens with the author's sentence 'We omit the proof of the implications (1) => (2) => (3).'; that sentence is kept verbatim and no step was supplied in its place. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The cut-complex statement relating the cohomology of the derived fibre to that of the complex. invoked at proof lines 20050-20051 and 20055-20059. Converts the residue field conditions into vanishing statements for cohomology and then into the existence of a localizing element.
- Vanishing of a module can be checked on localizations at maximal ideals. invoked at proof lines 20052-20053. Upgrades the local vanishing H^i(K^bullet)_m = 0 to global vanishing.
- The gluing statements for perfectness and for tor amplitude. invoked at proof lines 20059-20060. Assemble the local pieces into the global conclusion.

Source reference. Original lemma 00HN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 649–649. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 19867: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19867–19867. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 19077: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19077–19077. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 17210: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17210–17210. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $a, b \in \mathbf{Z}$. Let $K^\bullet$ be a pseudo-coherent complex of R-modules. The following are equivalent*

(1) *$K^\bullet$ is perfect with tor amplitude in $[a, b]$,*
(2) *for every prime $\mathfrak{p}$ we have $H^i(K^\bullet \otimes^{\mathbf{L}}_R \kappa(\mathfrak{p})) = 0$ for all $i \notin [a, b]$, and*
(3) *for every maximal ideal $\mathfrak{m}$ we have $H^i(K^\bullet \otimes^{\mathbf{L}}_R \kappa(\mathfrak{m})) = 0$ for all $i \notin [a, b]$.*

Proof. We omit the proof of the implications (1) $\Rightarrow$ (2) $\Rightarrow$ (3). Assume (3). Let $i \in \mathbf{Z}$ with $i \notin [a, b]$. By Lemma 19867 we see that the assumption implies that $H^i(K^\bullet)_\mathfrak{m} = 0$ for all maximal ideals of R. Hence $H^i(K^\bullet) = 0$, see Algebra, Lemma 00HN. Moreover, Lemma 19867 now also implies that for every maximal ideal $\mathfrak{m}$ there exists an element $f \in R$, $f \notin \mathfrak{m}$ such that $K^\bullet \otimes_R R_f$ is perfect with tor amplitude in $[a, b]$. Hence we conclude by appealing to Lemmas 19077 and 17210. □